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PROFESSIONAL BACKGROUND AND EDUCATION

April 2025–Present	Indian Institute of Science Education and Research Mohali, India Assistant Professor, Department of Biological Sciences Research area: Molecular Ecology and ancient DNA
Feb 2021–April 2025	Ashoka University, Sonapat, India Faculty Fellow, Ramalingaswami Fellow, Centre for Interdisciplinary Archaeological Research and Department of Biology Research area: Sexual selection and ancient DNA
Aug 2020–Jan 2021	Institute of Bioinformatics and Applied Biotechnology, Bangalore, India Ramanujan Fellow Research area: Speciation genomics
Dec 2014–Sept 2019	National University of Singapore, Singapore Research fellow, Department of Biological Sciences Research areas: Population genomics, Phylogenomics, Biogeography, and Speciation
Aug 2008–Aug 2014	National Centre for Biological Sciences, Tata Institute of Fundamental Research, Bangalore, India Ph.D., Biology, Advisor: Prof. Uma Ramakrishnan Thesis: Mating games: lessons from harem-forming fruit bat, <i>Cynopterus sphinx</i>
June 2006–April 2008	Madurai Kamaraj University, Madurai, India M.Sc. Genomics. <u>Gold medalist</u>
June 2003–April 2006	Bangalore University, Bangalore, India B.Sc. Biochemistry, Genetics, Biotechnology. <u>Gold medalist</u>

RESEARCH FUNDING

Active

Title: Sexual selection in birds: a genomic perspective

Funding agency: Department of Biotechnology, India

Role: PI

Grant period: 2022–2026

Title: Exaggerated male traits: understanding the genomics basis of sexual dimorphism in pheasants

Funding agency: Ramalingaswami fellowship, Department of Biotechnology, India

Role: PI

Grant period: 2021–2026

Title: Impact of climate change on small mammal communities and risks of spillover infections

Funding agency: Centre for Climate Change and Sustainability, Ashoka University.

Role: Co-PI

Grant period: 2022–2023

Previous

Title: Understanding speciation through the lens of genomics of reproductive isolation

Funding agency: Ramanujan Fellowship, Department of Science and Technology.

Role: PI

Grant period: 2020–2025 (Declined in 2021 to switch to Ramalingaswami Fellowship)

PUBLICATIONS

1. Dovih DP, Karjee R, **Garg KM**, Chattopadhyay B. 2025. Northeast India as the Next Frontier of Climate Change and One Health Research. *Ecohealth*, accepted.
2. Karjee R, Iyer V, Chatterjee D, Ray R, **Garg KM**, Chattopadhyay B[#]. 2025. Species biology and demographic history determines species vulnerability to climate change in tropical island endemic birds. *eLife*14:RP106369, under revision
3. **Garg KM**, Low DH, Lee G, Lee BP, Mendenhall IH, Rheindt FE, Smith GJ[#], Chattopadhyay B[#]. 2025. Hybrid genome assembly of the widespread bat *Rhinolophus lepidus* provides insights into susceptibility to SARS-CoV-2 infection and climate change threat. *DNA Research*. 32(3):dsaf015.
4. Ansil BR, Sanyal A, Sreenivas D, **Garg KM**, Ramakrishnan U, Chattopadhyay B. 2025. Identification of distinct rodent-associated adenovirus lineages from mixed-use landscape. *EcoHealth*, 1–6.
5. **Garg KM**[#], Dovih DP, Chattopadhyay B. Hybrid *de novo* genome assembly of the sexually dimorphic Lady Amherst's pheasant. **DNA Research**; 31:dsae001 ([#]Corresponding author) <https://doi.org/10.1093/dnares/dsae001>
6. Khan A, Sil M, Thekaekara T, **Garg KM**, Sinha I, Khurana R, Sukumar R, Ramakrishnan U. Divergence and serial colonization shape genetic variation and define conservation units in Asian elephants. **Current Biology**, 34, 1–12. [10.1016/j.cub.2024.08.062](https://doi.org/10.1016/j.cub.2024.08.062)
7. Lamba V, Herlekar I, Chatterjee D, Patel K, **Garg KM**, Chattopadhyay B. 2024. Signature of viral fossils: a comparative genomics approach to understand the diversity of endogenous retroviruses in bat. **Virus Research**, 350, 199484. <https://doi.org/10.1016/j.virusres.2024.199484>
8. Luo W, Tang Q, Chattopadhyay B, **Garg KM**, Rheindt FE, Wee AKS. 2024. The last stand: demographic and population genomic analysis reveals terminal endangerment in tropical timber species *Vatica guangxiensis*. **Conservation Letters**, 17:e13036.
9. **Garg KM**, Gwee CY, Chattopadhyay B, Ng NS, Prawiradilaga DM, David G, Fuchs J, Manh HL, Martinez J, Olsson U, Tu VT, Chhin S, Alström, P, Lei F, Rheindt FE. 2024. When colors mislead: genomics and bioacoustics prompt re-classification of Asian flycatcher radiation (Aves: Niltavinae). **Molecular Phylogenetics and Evolution**, 193, <https://doi.org/10.1016/j.ympev.2023.107999>.
10. Tan HZ, Jansen JJ, Allport GA, **Garg KM**, Chattopadhyay B, Irestedt M, Chilton G, Gwee CY, Rheindt FE. 2023. Megafaunal extinctions, not climate change, may explain Holocene genetic diversity declines in *Numenius* shorebirds. **eLife**, 12:e85422.
11. **Garg KM**, Lamba V, Chattopadhyay B. 2023. Genomic insights into the evolution and demographic history of the SARS-CoV-2 Omicron variant: population genomics approach. **JMIR Bioinformatics and Biotechnology**, 4:e40673.

12. **Garg KM**, Lamba V, Sanyal A, Dovih P, Chattopadhyay B[#]. 2023. Next generation sequencing revolutionizes organismal biology research in bats. **Journal of Molecular Evolution**, 91:391–404.
13. Ng EY, Li S, Zhang D, **Garg KM**, Song G, Martinez J, Hung LM, Tu VT, Fuchs J, Dong L, Olsson U. 2023. Genome-wide SNPs confirm plumage polymorphism and hybridisation within a *Cyornis* flycatcher species complex. **Zoologica Scripta**, 52:1–6. <https://doi.org/10.1111/zsc.12568>
14. **Garg KM**[#], Chattopadhyay B, Cros E, Tomassi S, Bennedick S, Edwards DP, Rheindt FE[#]. 2022. Island biogeography revisited: Museomics reveals affinities of shelf island birds determined by bathymetry and paleo-rivers, not by distance to mainland. **Molecular Biology and Evolution**, 39: msab340. <https://doi.org/10.1093/molbev/msab340> ([#]Corresponding authors)
15. Sanyal A, Agarwal S, Ramakrishnan U, **Garg KM**, Chattopadhyay B. 2022. Using environmental sampling to enable zoonotic pandemic preparedness. **Journal of the Indian Institute of Science**, 102:711–730. <https://doi.org/10.1007/s41745-022-00322-z>
16. **Garg KM**, Chattopadhyay B. 2021. Gene Flow in Volant Vertebrates: Species Biology, Ecology and Climate Change. **Journal of the Indian Institute of Science**, 101:165–176. <https://doi.org/10.1007/s41745-021-00239-z>
17. Bavejaa P, **Garg KM**[†], Chattopadhyay B[†], Sadanandan KR, Prawiradilaga DM, Yuda P, Lee JGH, Rheindt FE. 2021. Using historical genome-wide DNA to unravel the confused taxonomy in a songbird lineage that is extinct in the wild. **Evolutionary Applications**, 14:698–709. ([†]equal contribution) <https://doi.org/10.1111/eva.13149>
18. Chattopadhyay B, **Garg KM**, Swami Doss DP, Vinothkumar AK, Kandula S, Rheindt FE, Ramakrishnan U. 2021. Cryptic diversity of *Rhinolophus lepidus* in South Asia and differentiation across a biogeographic barrier. **Frontiers of Biogeography**, 13. <http://dx.doi.org/10.21425/F5FBG49625>
19. Chattopadhyay B, Forcina G, **Garg KM**, Irestedt M, Guerrini M, Barbanera F, Rheindt FE. 2021. Novel genome reveals susceptibility of popular gamebird, the red-legged partridge (*Alectoris rufa*, Phasianidae), to climate change. **Genomics**, 113:3430–3438. <https://doi.org/10.1016/j.ygeno.2021.08.010>
20. **Garg KM**, Chattopadhyay B, Koane B, Sam K, Rheindt FE. 2020. Last Glacial Maximum led to community-wide population expansion in a montane songbird radiation in highland Papua New Guinea. **BMC Evolutionary Biology**, 20:82. <https://doi.org/10.1186/s12862-020-01646-z>
21. Gwee CY, **Garg KM**[†], Chattopadhyay B[†], Sadanandan KR, Prawiradilaga DM, Irestedt M, Lei F, Lee JGH, Irham M, Haryoko T, Rowe KMC, Ferasyi TR, Wu S, Wogan G, Bowie RCK, Rheindt FE. 2020. Phylogenomics of white-eyes, a ‘great speciator’, reveals Indonesian archipelago as the center of lineage diversity. **eLife**, 9:e62765. ([†]equal contribution) <https://doi.org/10.7554/eLife.62765>
22. Chattopadhyay B, **Garg KM**, Ray R, Mendenhall IH, Rheindt FE. 2020. Novel *de novo* genome of *Cynopterus brachyotis* reveals evolutionarily abrupt shifts in gene family composition across fruit bats. **Genome Biology and Evolution**, 12:259–272. <https://doi.org/10.1093/gbe/evaa030>
23. Shingate P, Ravi V, Prasad A, Tay BH, **Garg KM**, Chattopadhyay B, Yap LM, Rheindt FE, Venkatesh B. 2020. Chromosome-level assembly of the horseshoe crab genome provides insights into its genome evolution. **Nature Communications**, 11:2322. <https://doi.org/10.1038/s41467-020-16180-1>

24. Cros E, Chattopadhyay B, **Garg KM**, Ng NS, Tomassi S, Benedick S, Edwards DP, Rheindt FE. 2020. Quaternary land bridges have not been universal conduits of gene flow. **Molecular Ecology**, 29:2692–2706. <https://doi.org/10.1111/mec.15509>
25. **Garg KM**, Sam K, Chattopadhyay B, Koane B, Ericson PGP, Rheindt FE. 2019. Secondary gene flow in the Müllerian mimicry ring of a poisonous Papuan songbird clade (*Pitohui*; Aves). **Genome Biology and Evolution**, 11:2332–2343. <https://doi.org/10.1093/gbe/evz168>
26. Chattopadhyay B, **Garg KM**, Mendenhall IH, Rheindt FE. 2019. Historic DNA reveals Anthropocene threat to a tropical urban fruit bat. **Current Biology**, 29:R1269–R1300. <https://doi.org/10.1016/j.cub.2019.11.013>
27. Chattopadhyay B, **Garg KM**, Jing SY, Low GW, Frechette J and Rheindt FE. 2019. Conservation genomics in the fight to help the recovery of the critically endangered Siamese crocodile *Crocodylus siamensis*. **Molecular Ecology**, 28:936–950. <https://doi.org/10.1111/mec.15023>
28. Chattopadhyay B, **Garg KM**, Ray R, Rheindt FE. 2019. Fluctuating fortunes: genomes and habitat reconstructions reveal global climate-mediated changes in bats' genetic diversity. **Proceedings of the Royal Society B**, 286:20190304. <https://doi.org/10.1098/rspb.2019.0304>
29. Gwee CY, Eaton JA, **Garg KM**, Alström P, Van Balen S, Hutchinson RO, Prawiradilaga DM, Le MH, Rheindt FE. 2019. Cryptic diversity in *Cyornis* (Aves: Muscicapidae) jungle-flycatchers flagged by simple bioacoustic approaches. **Zoological Journal of the Linnean Society**, 186:725–741. <https://doi.org/10.1093/zoolinnean/zlz003>
30. **Garg KM**, Chattopadhyay B, Wilton PR, Prawiradilaga DM, Rheindt FE. 2018. Pleistocene land bridges act as semipermeable agents of avian gene flow in Wallacea. **Molecular Phylogenetics and Evolution**, 125:196–203. <https://doi.org/10.1016/j.ympev.2018.03.032>
31. Tan DJ, Chattopadhyay B, **Garg KM**, Cros E, Ericson PG, Irestedt M, Rheindt FE. 2018. Novel genome and genome-wide SNPs reveal early fragmentation effects in an edge-tolerant songbird population across an urbanized tropical metropolis. **Scientific Reports**, 8:12804. <https://doi.org/10.1038/s41598-018-31074-5>
32. Low GW, Chattopadhyay B[†], **Garg KM**[†], Irestedt M, Ericson PG, Yap G, Tang Q, Wu S, Rheindt FE. 2018. Urban landscape genomics identifies fine-scale gene flow patterns in an avian invasive. **Heredity**, 120:138–153. ([†] equal contribution) <https://doi.org/10.1038/s41437-017-0026-1>
33. **Garg KM**[#], Chattopadhyay B and Ramakrishnan U. 2018. Social structure of the harem-forming promiscuous fruit bat, *Cynopterus sphinx*, is the harem truly important? **Royal Society Open Science**, 172024. ([#] Corresponding author) <https://doi.org/10.1098/rsos.172024>
34. **Garg KM**[#], Ramakrishnan U. 2017. Variance in Female Reproductive Success Differentially Impacts Effective Population Size in the Short-Nosed Fruit Bat, *Cynopterus sphinx*. **Evolutionary Biology**, 1–8. ([#] Corresponding author) <https://doi.org/10.1007/s11692-017-9414-y>
35. Chattopadhyay B, **Garg KM**, Gwee CY, Edwards SV, Rheindt FE. 2017. Gene flow during glacial habitat shifts facilitates character displacement in a Neotropical flycatcher radiation. **BMC Evolutionary Biology**, 17:1–5. <https://doi.org/10.1186/s12862-017-1047-3>
36. Ng NS, Wilton PR, Prawiradilaga DM, Tay YC, Indrawan M, **Garg KM**[#], Rheindt FE[#]. 2017. The effects of Pleistocene climate change on biotic differentiation in a montane songbird clade from Wallacea. **Molecular Phylogenetics and Evolution**, 114:353–366. ([#] Corresponding authors) <https://doi.org/10.1016/j.ympev.2017.05.007>

37. Çilingir FG, Rheindt FE, **Garg KM**, Platt K, Platt SG, Bickford DP. 2017. Conservation genomics of the endangered Burmese roofed turtle. **Conservation Biology**, 31:1469–1476. <https://doi.org/10.1111/cobi.12921>
38. Ng EY, **Garg KM**, Low GW, Chattopadhyay B, Oh RR, Lee JG, Rheindt FE. 2017. Conservation genomics identifies impact of trade in a threatened songbird. **Biological conservation**, 214:101-108. <https://doi.org/10.1016/j.biocon.2017.08.007>
39. Mani RS, Dovih DP, Ashwini MA, Chattopadhyay B, Harsha PK, **Garg KM**, Sudarshan S, Puttaswamaiah R, Ramakrishnan U, Madhusudana SN. 2017. Serological evidence of lyssavirus infection among bats in Nagaland, a north-eastern state in India. **Epidemiology & Infection**, 145:1635–1641. <https://doi.org/10.1017/S0950268817000310>
40. **Garg KM**, Tizard R, Ng NSR, Cros E, Dejtaradol A, Chattopadhyay B, Pwint N, Päckert M, Rheindt FE. 2016. Genome-wide data help identify an avian species-level lineage that is morphologically and vocally cryptic. **Molecular Phylogenetics and Evolution**, 102: 97–103. <https://doi.org/10.1016/j.ympev.2016.05.028>
41. Chattopadhyay B, **Garg KM**, Vinoth Kumar AK, SwamiDoss DP, Rheindt FE, Kandula S and Ramakrishnan U. 2016. Genome-wide data reveal cryptic diversity and genetic introgression in an Oriental cynopterine fruit bat radiation. **BMC Evolutionary Biology**, 16:1. <https://doi.org/10.1186/s12862-016-0599-y>
42. **Garg KM**[#], Chattopadhyay B, SwamiDoss PS, Vinoth Kumar AK, Kandula S and Ramakrishnan U. 2015. Benefits of sociality: Males and females gain differentially from sociality in a promiscuous fruit bat *Cynopterus sphinx*. **PLoS ONE**, 10(3): e0122180. ([#] Corresponding author) <https://doi.org/10.1371/journal.pone.0122180>
43. Chattopadhyay B, **Garg KM**, Ramakrishnan U. 2014. Effect of diversity and missing data on genetic assignment with RAD-Seq markers. **BMC Research Notes**, 7:1–3. <https://doi.org/10.1186/1756-0500-7-841>
44. **Garg KM**[#], Chattopadhyay B, SwamiDoss PS, Vinoth Kumar AK, Kandula S and Ramakrishnan U. 2012. Promiscuous mating in a harem bat *Cynopterus sphinx*. **Molecular Ecology**, 21: 4092–4105. ([#] Corresponding author) <https://doi.org/10.1111/j.1365-294X.2012.05665.x>
45. Chattopadhyay B, **Garg KM**, Vinoth Kumar AK, Ramakrishnan U, Kandula S. 2012. Sibling species in South Indian populations of the rufous horse-shoe bat *Rhinolophus rouxii*. **Conservation Genetics**, 13:1435–1445. <https://doi.org/10.1007/s10592-012-0361-y>
46. Chattopadhyay B, Schuller G, **Garg KM**, Kandula S. 2010. A new phonic type of the rufous horseshoe bat *Rhinolophus rouxii* from southern India. **Current Science**, 10:114–118.

#Corresponding author

Google Scholar Profile:

<https://scholar.google.com.sg/citations?user=NaxWa4oAAAAJ&hl=en>

BOOK CHAPTERS

1. **Garg KM**[#], Chattopadhyay B, SwamiDoss DP, VinothKumar AK, Kandula S. 2023. Bat societies across habitat types: insights from a commonly occurring fruit bat *Cynopterus sphinx*. In: Moretto L, Fenton B, Patriquin K, Coleman J, Korine C, Davy C (eds.) **Urban Bats - Biology, Ecology, and Human Dimensions**. Springer Nature, Switzerland.
2. Chattopadhyay B, **Garg KM**, Ray R, Mendenhall IH, Rheindt FE. 2023. Genetic impoverishment in the Anthropocene: A tale from bats. In: Moretto L, Fenton B, Patriquin K,

POPULAR SCIENCE ARTICLE

1. **Garg KM**. 2019. Last Ice Age: understanding Earth's climatic history. CEiBa Newsletter. Volume 2, Issue 1: 1–14. DOI: 10.13140/RG.2.2.11159.42409

CONFERENCE PRESENTATIONS

1. **Garg KM**, Chattopadhyay B, Cros E, Tomassi S, Bennedick S, Edwards DP, Rheindt FE. Impact of Quaternary glacial cycles on avifauna in Southeast Asia. 3rd AsiaEvo Meeting, National University of Singapore, 2023.
2. **Garg KM**. Impact of Pleistocene glacial cycles on avifauna in Southeast Asia and Australasia. Fifth Bangalore School on Population Genetics and Evolution, 2021. Invited speaker.
3. **Garg KM**, Chattopadhyay B, Wilton P, Dewi IP, Rheindt FE. Pleistocene land bridges act as semipermeable agents of avian gene flow in Wallacea. International Biogeography Society India Meeting, 2017, Bangalore, India.
4. **Garg KM**, Chattopadhyay B and Ramakrishnan U. Paternity in the NGS era. Evolution, 2014, Raleigh, USA.
5. **Garg KM**, Chattopadhyay B, Vinoth Kumar AK, SwamiDoss DP, Kandula S and Ramakrishnan U. Post-copulatory mate choice for heterozygous offspring in harem forming fruit bat, *Cynopterus sphinx*. Evolution, 2013, Snowbird, Salt Lake City, USA.
6. **Garg KM**, Chattopadhyay B, Vinoth Kumar AK, SwamiDoss DP, Kandula S and Ramakrishnan U. Promiscuous mating in the harem-roosting fruit bat, *Cynopterus sphinx*. 12th International Society for Behavioral Ecology, 2012, Lund, Sweden.
7. **Garg KM**, Chattopadhyay B, Vinoth Kumar AK, SwamiDoss DP, Kandula S and Ramakrishnan U. Is Harem Size a Predictor of Male Reproductive Success Across Habitats and Seasons? 15th International Bat Research Conference, 2010, Prague.

AWARDS/ACHIEVEMENTS

- Society for the Study of Evolution Travel Award, 2013.
- Best poster award along with travel award National Centre for Biological Sciences, 2013.
- Travel award to attend 12th International Society for Behavioral Ecology, 2012.
- Best student award from National Centre for Biological Sciences, 2009.
- 'Maveeran Sundaralingam Endowment Scholarship' for having secured the highest marks in M.Sc., 2008.
- First in Bangalore University B.Sc. Vocational Sciences, 2007.
- Scholarship by the Chairman of the Jain Group of Institutes for B.Sc.
- All India Rank 253 in Graduate Aptitude Test in Engineering (GATE), 2007.
- Best Student Award from Sri Bhagawan Mahaveer Jain College, 2007.
- Certificate for the Academic Excellence from Sri Bhagawan Mahaveer Jain College, 2006.